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1. (a) Show that

$$3 \cosh x - \sinh x = e^x + 2e^{-x}$$

[2]

(b) Hence solve the differential equation

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 6(3 \cosh x - \sinh x)$$

given that, when $x = 0$, $y = 1$ and $\frac{dy}{dx} = 1$.

[10]

2. Find the particular solution of the differential equation

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 13y = 8e^{-2t} \sin t$$

given that, when $t = 0$, $y = 1$ and $\frac{dy}{dt} = 2$.

[10]

3. A damped oscillator is governed by

$$\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 13y = 75 \sin 2t$$

(a) Find the general solution of this differential equation. [6]

(b) Show that, whatever the initial conditions, the motion is approximately of the form

$$y \approx R \sin(2t - \phi)$$

for large positive values of t , where $R > 0$ and $0 < \phi < \frac{\pi}{2}$. Determine R and ϕ . [4]

4. A function $y(t)$ satisfies the differential equation

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 5y = 10 \cos t$$

Given that, when $t = 0$, $y = 5$ and $\frac{dy}{dt} = 2$, find the particular solution.

[10]

5. (a) Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 13y = 0 \quad [2]$$

(b) Hence find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 13y = e^{-2x}(6 \cos x - 8 \sin x) \quad [4]$$

6. (a) Show that

$$\frac{\cosh x - \sinh x}{\cosh x + \sinh x} = e^{-2x}$$

[2]

(b) Hence find the particular solution of the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 4 \left(\frac{\cosh x - \sinh x}{\cosh x + \sinh x} \right)$$

given that, when $x = 0$, $y = 1$ and $\frac{dy}{dx} = 1$.

[10]

7. Solve the differential equation

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 50 \sin x$$

subject to the conditions $y(0) = 3$ and $y'(0) = -6$.

[10]

8. Determine the solution $y(x)$ of the initial value problem

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = xe^{2x}$$

subject to

$$y(0) = 1, \quad \left. \frac{dy}{dx} \right|_{x=0} = 0 \quad [10]$$

9. (a) Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 4y = 0 \quad [2]$$

- (b) Hence, by considering a particular integral of the form $x(A \cos 2x + B \sin 2x)$, find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 4y = 8 \cos 2x - 12 \sin 2x \quad [4]$$

10. Find the particular solution of the differential equation

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 13y = 28 \cos 3t - 4 \sin 3t$$

given that, when $t = 0$, $y = 5$ and $\frac{dy}{dt} = 7$.

[10]

11. Find the particular solution of the differential equation

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 5y = 8\cos t$$

given that, when $t = 0$, $y = 3$ and $\frac{dy}{dt} = 1$.

[10]

12. (a) Show that, for all real x ,

$$(1 + \tanh x) \cosh x = e^x \quad [2]$$

(b) Hence solve the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 3(1 + \tanh x) \cosh x$$

given that, when $x = 0$, $y = 1$ and $\frac{dy}{dx} = 0$. [10]